# Supporting information for: Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

This supporting information accompanies the manuscript titled *Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling* and provides expanded workflow notes and supplemental derivations.

## S1 Staged refinement workflow (expanded)

1. **Beam characterization (direct beam).** Acquire direct-beam images at several detector distances. Fit the spot position and shape versus distance to determine the incident-beam widths $(w_x, w_z)$ and Gaussian divergences $(\Delta\theta_x, \Delta\theta_z)$ that set the instrument-limited broadening.

2. **Detector geometry calibration (3D powder standard).** Measure a polycrystalline hBN calibrant and subtract the detector dark frame. Fit multiple Debye–Scherrer rings in detector coordinates and refine detector distance, detector tilts $(\beta, \gamma)$, in-plane detector rotation $\chi$, and beam center $(x_0, y_0)$.

3. **Sample alignment and goniometer misalignment.** For each dark-subtracted sample image, use indexed Bragg-peak centers rather than full-image intensity. With detector parameters fixed, refine sample alignment by minimizing detector-space residuals between measured and simulated peak positions spanning both specular and off-specular regions. Refine $(\theta_i, \delta, z_B, z_S)$ and the goniometer-axis misalignment rotations $(\alpha, \psi)$; include film dimensions $(W, H)$ and thickness $t$ to weight footprint and absorption.

4. **Mosaicity/profile refinement.** With geometry fixed, compare selected local detector ROIs or local $I(\phi)$ profiles after center alignment. The shared profile model is a wrapped Gaussian-plus-Lorentzian angular distribution, not a pseudo-Voigt common-width peak approximation. Off-specular arcs mainly constrain the Gaussian core, while off-condition specular intensity constrains the Lorentzian tail. Report Lorentzian FWHM, Gaussian FWHM, and the relative Lorentzian weight.

5. **Ordered-structure refinement.** With geometry and mosaicity fixed, refine the ordered intensity model against background-subtracted integrated counts in selected detector-space Bragg ROIs from multiple dark-subtracted images. The corresponding prediction is the simulated detector-carried intensity deposited in those same ROIs. Use one nuisance scale per image and physically constrained ordered-structure parameters.

6. **Disorder refinement.** After the ordered model converges, introduce stacking-disorder parameters only through selected $Q_r/Q_z$ rod profiles. The measured profile is formed from caked signal and normalization fields by summing signal and normalization over the selected rod mask before division. Geometry, detector calibration, beam terms, mosaicity, lattice constants, and the ordered baseline remain fixed; only stacking-disorder parameters and declared scale/background terms are refined.

All detector images are dark-subtracted before fitting. Any additional preprocessing (flat-field, polarization, solid-angle, or detector-efficiency corrections) should either be applied consistently to all images before refinement or stated explicitly if omitted. We recommend stage-specific uncertainty estimation by repeated constrained refits: resample the calibrant points for detector geometry, indexed peak lists for sample alignment, local reflection regions for mosaicity, accepted Bragg ROIs for ordered intensity, and selected rod bins for disorder. Report the standard deviation of the accepted solutions and propagate parameters to the next stage only after the preceding stage is stable.

## S1.1 Wrapped mosaic distribution

In the main text we describe out-of-plane mosaicity by a misorientation density $\omega(\Delta\theta)$ in the crystallite tilt angle $\Delta\theta$ relative to the substrate normal. Since $\Delta\theta$ is an orientation angle, $\Delta\theta$ and $\Delta\theta + 2\pi$ represent the same physical state. Accordingly, $\omega$ is treated as a $2\pi$-periodic probability density on $\Delta\theta \in (-\pi, \pi]$.

The implemented profile is a wrapped Gaussian plus a wrapped Lorentzian,

$$\omega_{\mathrm{w}}(\Delta\theta; \eta, \Gamma, \sigma) = \eta\, L_{\mathrm{w}}(\Delta\theta; \Gamma) + (1-\eta)\, G_{\mathrm{w}}(\Delta\theta; \sigma), \qquad 0 \le \eta \le 1.$$

This is not a pseudo-Voigt constraint: the Lorentzian half-width $\Gamma$ and Gaussian standard deviation $\sigma$ are independent. The wrapped Lorentzian and wrapped Gaussian are

$$L_{\mathrm{w}}(\Delta\theta; \Gamma) = \sum_{m=-\infty}^{\infty} \frac{1}{\pi} \frac{\Gamma}{(\Delta\theta + 2\pi m)^2 + \Gamma^2} = \frac{1}{2\pi} \frac{\sinh \Gamma}{\cosh \Gamma - \cos \Delta\theta},$$

$$\begin{aligned} G_{\mathrm{w}}(\Delta\theta; \sigma) &= \sum_{m=-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(\Delta\theta + 2\pi m)^2}{2\sigma^2}\right] \\ &= \frac{1}{2\pi}\left[1 + 2\sum_{n=1}^{\infty} e^{-n^2\sigma^2/2} \cos(n\Delta\theta)\right]. \end{aligned}$$

Both components integrate to unity over $(-\pi, \pi]$, so $\omega_{\mathrm{w}}$ is normalized. The reported Lorentzian FWHM is $2\Gamma$, and the reported Gaussian FWHM is $2\sqrt{2\ln 2}\,\sigma$. The local small-width limit recovers the usual line-shape widths, but the refinement and normalization are defined by the wrapped functions above.

For a reflection with Bragg-sphere radius $G \equiv |\mathbf{G}|$, mosaicity redistributes intensity in polar angle $\vartheta$ with line density $I_0\, \omega_{\mathrm{w}}(\vartheta - \vartheta_B)\, d\vartheta$, where $\vartheta_B$ is the Bragg position for that reflection. Random in-plane rotations about the film normal revolve this line density about the $G_z$ axis, yielding an axially symmetric surface density $\rho(\vartheta)$. With area element $dA = 2\pi G^2 \sin\vartheta\, d\vartheta$, conservation of intensity gives

$$\rho(\vartheta) = \frac{I_0}{2\pi G^2 \sin\vartheta}\, \omega_{\mathrm{w}}(\vartheta - \vartheta_B), \qquad 0 < \vartheta < \pi,$$

as used in the main text.

## S1.2 Wave propagation, refraction, and absorption

Wave attenuation occurs both at the interfaces (Fresnel transmission) and as the wave propagates through the film (absorption). In the forward model we therefore (i) refract each ray into and out of the film, (ii) assign the absorption to the out-of-plane component of the internal wavevector, and (iii) weight scattering from a finite thickness $t$ by thickness-averaged Beer–Lambert attenuation. This subsection records the explicit equations used.

**Complex refractive index.** We write the complex refractive index as

$$\tilde{n} = 1 - \delta_n + i\beta_n, \tag{S1}$$

where the subscript $n$ avoids confusion with the geometric angles $(\delta, \beta)$ used elsewhere. In terms of the wavelength $\lambda$, electron density $\rho_e$, classical electron radius $r_e$, and linear absorption coefficient $\mu$,

$$\delta_n = \frac{r_e \lambda^2 \rho_e}{2\pi}, \qquad \beta_n = \frac{\mu\lambda}{4\pi}. \tag{S2}$$

**Angles and refraction.** Let $k_0 = 2\pi/\lambda$ and take $\hat{z}$ along the sample normal. For an incident ray in air, the grazing-incidence angle $\theta_i$ and in-plane azimuth $\varphi_i$ may be expressed in terms of the external wavevector components as

$$\sin\theta_i = \frac{k_{i,z}}{k_0}, \tag{S3}$$

$$\tan\varphi_i = \frac{k_{i,x}}{k_{i,y}}. \tag{S4}$$

Upon refraction, we compute the transmitted grazing angle using the real part of the refractive index,

$$\theta_t = \cos^{-1}\left(\frac{\cos\theta_i}{\mathrm{Re}(\tilde{n})}\right). \tag{S5}$$

**Internal wavevector components.** We define a real internal wavevector magnitude for the in-plane components as

$$k' = k_0 \left(\mathrm{Re}\left(\tilde{n}^2\right)\right)^{1/2}, \tag{S6}$$

so that

$$k_{tx} = k' \cos\theta_t \sin\varphi_i, \tag{S7}$$

$$k_{ty} = k' \cos\theta_t \cos\varphi_i. \tag{S8}$$

To satisfy the boundary condition at the surface, we take only the out-of-plane component to be complex and write

$$k_{tz}^2 = a + ib, \qquad k_{tz} = u + iv = \sqrt{a + ib}. \tag{S9}$$

Expanding $(u + iv)^2$ gives

$$(u + iv)^2 = u^2 - v^2 + 2iuv = a + ib,$$

so that the real and imaginary parts satisfy

$$a = u^2 - v^2 = (u+v)(u-v), \qquad b = 2uv = (u+v)^2 - u^2 - v^2. \tag{S10}$$

The magnitude follows from

$$|k_{tz}|^2 = u^2 + v^2 = \sqrt{(a+ib)(a-ib)} = \sqrt{a^2+b^2}, \tag{S11}$$

which yields

$$u^2 = \frac{\sqrt{a^2+b^2}+a}{2}, \qquad v^2 = \frac{\sqrt{a^2+b^2}-a}{2}. \tag{S12}$$

In our parameterization,

$$\mathrm{Re}(k_{tz}^2) = k'^2 \sin^2\theta_t \equiv a_t, \tag{S13}$$

$$\mathrm{Im}(k_{tz}^2) = k_0^2\ \mathrm{Im}\left(\tilde{n}^2\right) \equiv b, \tag{S14}$$

so that

$$\mathrm{Re}(k_{tz}) = \sqrt{\frac{\sqrt{a_t^2+b^2}+a_t}{2}}, \tag{S15}$$

$$\mathrm{Im}(k_{tz}) = \sqrt{\frac{\sqrt{a_t^2+b^2}-a_t}{2}}. \tag{S16}$$

We take the physical branch $\mathrm{Im}(k_{tz}) > 0$ so that the field decays into the film.

**Exit path and refraction out of the film.** For the scattered ray we repeat the same construction using the external grazing-exit angle and in-plane azimuth $(\theta_f, \varphi_f)$. We compute an internal exit angle $\theta_t'$ (again using $\mathrm{Re}(\tilde{n})$) and form

$$k_{tx}' = k' \cos\theta_t' \sin\varphi_f, \tag{S17}$$

$$k_{ty}' = k' \cos\theta_t' \cos\varphi_f, \tag{S18}$$

$$a_f \equiv k'^2 \sin^2\theta_t', \tag{S19}$$

with the same $b_f = b_t = b$,

$$\mathrm{Re}(k_{tz}') = \sqrt{\frac{\sqrt{a_f^2+b^2}+a_f}{2}}, \tag{S20}$$

$$\mathrm{Im}(k_{tz}') = \sqrt{\frac{\sqrt{a_f^2+b^2}-a_f}{2}}. \tag{S21}$$

Optical reversibility then gives the refraction back into air; inverting the relation for $\theta_t$ gives

$$\theta_f = \cos^{-1}\left(\mathrm{Re}(\tilde{n})\cos\theta_t'\right). \tag{S22}$$

**Fresnel transmission factors.** We weight each ray by Fresnel transmission factors at entry and exit interfaces. For a scalar field written in terms of the normal wavevector components, one convenient form for an amplitude transmission coefficient from medium 1 to 2 is

$$t_{12} = \frac{2k_{1z}}{k_{1z} + k_{2z}}. \tag{S23}$$

**Absorption weight (uniform scattering through thickness).** Assuming a uniform probability of scattering across depth $z \in [0, t]$ and Beer–Lambert attenuation along the incident and exit paths, the intensity weight used in the forward model is

$$w_{\text{abs}} = \frac{1}{t} \int_0^t \exp\left[-\mu z \left(\frac{1}{\sin\theta_i} + \frac{1}{\sin\theta_f}\right)\right] dz \tag{S24}$$

$$= \frac{1 - \exp\left[-\mu t \left(\frac{1}{\sin\theta_i} + \frac{1}{\sin\theta_f}\right)\right]}{\mu t \left(\frac{1}{\sin\theta_i} + \frac{1}{\sin\theta_f}\right)}. \tag{S25}$$

## S1.3 Detector-derived ordered and disorder observables

The detector-space forward model predicts full images, but the inverse problems use stage-specific observables. In the ordered-structure stage, the measured quantity for ROI $i$ in image $m$ is the background-subtracted detector count

$$y_{mi} = \sum_{(u,v)\in\Omega_{mi}} \left[I_m^{\text{meas}}(u, v) - b_{mi}(u, v)\right],$$

where $\Omega_{mi}$ is the selected detector-space Bragg ROI and $b_{mi}$ is the local background model. The corresponding prediction is the simulated detector-carried intensity deposited in the same ROI,

$$\mu_{mi} = \sum_{j\in\mathcal{R}_{mi}} w_j,$$

with geometry, beam terms, and mosaicity fixed. One nuisance scale per image accounts for exposure and flux differences. This makes the ordered stage a multi-image ROI-integrated Bragg-intensity refinement, not a full-image pixel objective.

The $PbI_2$ disorder refinement is a final-stage selected-profile fit, not a second global image refinement. The upstream beam, detector, sample alignment, mosaic/profile parameters, lattice constants, and ordered $PbI_2$ baseline are fixed before this stage starts. The detector image is first caked into a signal field $S(\phi, 2\theta)$ and a normalization field $N(\phi, 2\theta)$ using the calibrated detector geometry. For the selected-rod conversion, let $\eta$ denote the polar angle of the scattering vector relative to the film normal. For each caked bin,

$$Q = \frac{4\pi}{\lambda} \sin\left(\frac{2\theta}{2}\right), \qquad Q_r = |\sin\eta| Q, \qquad Q_z = \cos\eta\, Q.$$

A selected rod with in-plane radius $Q_{r,0}$ is retained when

$$|Q_r - Q_{r,0}| \le \Delta Q_r, \qquad Q_{z,\min} \le Q_z \le Q_{z,\max}.$$

A branch selector may be applied after this reciprocal-space mask so that one branch is used for the default fit and the symmetry-related branch is retained for validation.

Let $S_{ij}$ and $N_{ij}$ denote the caked signal and normalization fields, and let $m_{ij}$ denote the selected-rod mask. The measured profile is

$$I_j^{\mathrm{rod}} = \frac{\sum_i m_{ij} S_{ij}}{\sum_i m_{ij} N_{ij}},$$

for bins with positive selected normalization. Signal and normalization are therefore summed before division. This prevents the final trace from becoming a sum of already normalized pixels and keeps the observable tied to the detector-space calibration.

The model profile is evaluated along the same selected branch. In compact notation,

$$M_j(\boldsymbol{\theta}_{\mathrm{dis}}) = B(x_j) + s\,[K * I_{hk}^{\mathrm{nc}}(L; \boldsymbol{\theta}_{\mathrm{dis}})]_j,$$

where $x_j$ is the profile coordinate, $L$ is the corresponding stacking coordinate on the selected branch, $B$ is a low-order background, $s$ is a global scale, $K$ is the fixed detector/profile broadening kernel, and $I_{hk}^{\mathrm{nc}}$ is the no-correlation analytical stacking intensity from Sec. S3. When an explicit closed-form broadening kernel is used, $K$ is written as a Gaussian-plus-Lorentzian kernel with widths fixed by the upstream resolution and mosaic/profile state or declared before the disorder fit. It is not used to release independent peak areas.

The disorder objective is evaluated only over the retained profile bins,

$$\chi^2_{\mathrm{dis}}(\boldsymbol{\theta}_{\mathrm{dis}}, s, B) = \sum_{j \in \mathcal{J}} \rho\left(\frac{I_j^{\mathrm{rod}} - M_j(\boldsymbol{\theta}_{\mathrm{dis}})}{\sigma_j}\right),$$

where $\rho$ may be a least-squares or robust loss and $\mathcal{J}$ is the retained $Q_z$ interval. The known Bragg locations on the selected branch are inherited from the fixed detector geometry and ordered model. Peak amplitudes are not fit independently before the stacking model is applied; otherwise the diffuse between-peak intensity would already have been absorbed by the preprocessing.

## S2 Material background and stacking–fault model

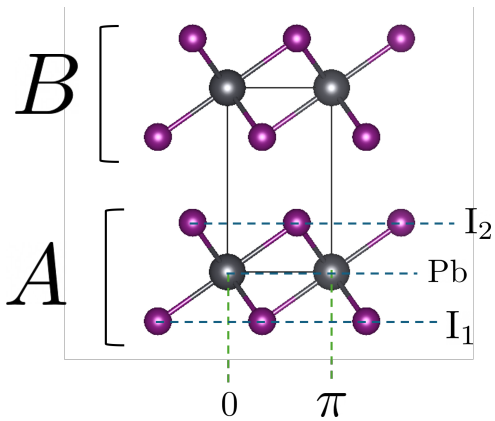
In layered metal–halides such as $PbI_2$ each I–Pb–I sandwich behaves like a rigid trilayer that can shift laterally with negligible energy cost compared with the much stronger intra-layer bonding. Consequently the crystal can adopt several polytypes that differ only in the stacking sequence of these sandwiches along the $c$ axis. Here we focus on the two sequences that bracket the experimentally relevant window:

### S2.1 Polytypism of $PbI_2$

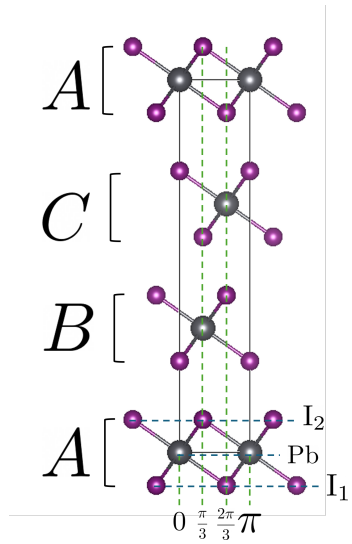
**2H ($\boldsymbol{P\bar{3}}$m1)** AB repeat stable below 140 °C; no lateral shift ($\phi_{AB} = 0$); see Fig. S1a.

**6H ($\boldsymbol{R\bar{3}}$)** ABCA repeat above $\approx 200$ °C; lateral shift $a/\sqrt{3}$ gives $\phi = 2\pi/3$; 2H $\Leftrightarrow$ 6H transition near 94 °C; see Fig. S1b.

Our samples never achieve perfect periodicity; stacks often contain finite 2H and 6H "domains" interspersed by stacking faults. To model the resulting diffuse scattering we adapt the Hendricks–Teller (HT) Markov formalism and follow the streamlined matrix notation introduced a few years later by Lifshitz, who showed how to re-express pair probabilities through the eigenvalues of a $3 \times 3$ transition matrix.




**(a)** 2H polytype (AB) with no in-plane phase shifts.




**(b)** 6H R$\bar{3}$ polytype (ABCA) with $\frac{1}{3}$-lattice translations giving $e^{i2\pi/3}$ phase steps.

**Figure S1:** Low-temperature 2H and high-temperature 6H stacking sequences in $PbI_2$. Arrows illustrate the lateral shifts responsible for the phase factors used in the HT–Lifshitz model.

### S2.2 Molecular structure factor for $PbI_2$

A single layer form factor is

$$F_m = b_{Pb} + b_I \left[ e^{2\pi i(H/3+2K/3+Lz/3)} + e^{2\pi i(2H/3+K/3-Lz/3)} \right] \tag{S26}$$

Where now the amplitude will be

$$A = F_m \sum_{R_p} e^{i\vec{Q}\cdot\vec{R}_p} \tag{S27}$$

Where $\vec{R}_p = m\vec{a} + n\vec{b}$ is the in-plane position vector, with $m, n \in \mathbb{Z}$. Then $\vec{Q}\cdot\vec{R}_p \to Hm + kn$, and the Bragg condition makes $H, K$ integers.

## S3 No-correlation stacking-fault model

Let $\tau_j \in \{0, 1\}$ indicate whether the $j$-th interface slips ($\tau_j = 1$) or not ($\tau_j = 0$). The structure factor of one layer is $F_0(h, k, \ell)$, and for a three-layer (ABC) repeat the per-layer phase advance is

$$\phi(h, k, \ell) = \frac{h + 2k}{3} + \ell/3, \tag{S28}$$

with $h, k, \ell$ in reciprocal-lattice units.
We now add the phase of each layer to the form factor from equation S27.

$$A = F_m \sum_{R_p} e^{i\vec{Q}\cdot\vec{R}_p} \sum_{\ell} e^{i\phi(HKL)\ell} \tag{S29}$$

Writing $e^{i\phi\ell} = \prod_{j=1}^{\ell} e^{i\phi}$

$$A = F_m \sum_{R_p} e^{i\vec{Q}\cdot\vec{R}_p} \prod_{\ell} e^{i\phi(HKL)\ell} \tag{S30}$$

### S3.1 Coherent amplitude of an $N$-layer stack

At layer $\ell$ the factor multiplying $F_0$ is

$$\left[1 - \tau_\ell + \tau_\ell e^{-i\phi}\right] e^{i\phi\ell},$$

so that

$$A = F_0 \sum_{\ell=0}^{N-1} \left[\prod_{j=0}^{\ell} (1 - \tau_j + \tau_j e^{-i\phi})\right] e^{i\phi\ell}. \tag{S31}$$

### S3.2 Uncorrelated faults

Assume the $\tau_j$ are i.i.d. with slip probability $p = \Pr(\tau_j = 1)$. Independence gives

$$\left\langle \prod_{j=0}^{\ell} (1 - \tau_j + \tau_j e^{-i\phi}) \right\rangle = \left[(1-p) + p\, e^{-i\phi}\right]^{\ell} \equiv f^{\,\ell}, \qquad f \equiv (1-p) + p\, e^{-i\phi}. \tag{S32}$$

### S3.3 Mean amplitude

Averaging (S31) and inserting (S32) gives the geometric series

$$\langle A \rangle = F_0 \sum_{\ell=0}^{N-1} f^{\ell} e^{i\phi\ell} = F_0 \frac{1 - (f\, e^{i\phi})^N}{1 - f\, e^{i\phi}}. \tag{S33}$$

### S3.4 Infinite-crystal limit

For any non-zero fault probability ($|f| < 1$),

$$\boxed{\lim_{N\to\infty} \langle A \rangle = F_0 \frac{f}{1 - f\, e^{i\phi}}} \tag{S34}$$

The perfect-crystal result is recovered as $f \to 1$.

### S3.5 Mean-square amplitude

The heights of sharp Bragg peaks: i.e. the square of the coherently averaged structure amplitude. This is only the part of the intensity that survives perfect phase coherence: sharp Bragg peaks. Any layer-to-layer fluctuations are averaged before squaring and thus disappear.

$$
\begin{aligned}
|\langle A\rangle|^2 &= |F_0|^2 \frac{f^2}{|1 - f\, e^{i\phi}|^2} \\
&= |F_0|^2 \frac{f^2}{(1 - f\cos\phi)^2 + f^2 \sin^2\phi} && \text{(S35)} \\
&= |F_0|^2 \frac{f^2}{(1-f)^2 + 4f\sin^2\frac{\phi}{2}}. && \text{(S36)}
\end{aligned}
$$

### S3.6 Ensemble-averaged intensity

The full detector signal; adds the broadened peaks and diffuse background arising from stacking faults.

$$< |A|^2 > \tag{S37}$$

#### S3.6.1 Layer Autocorrelation

The layer-resolved autocorrelation of the stacking sequence tells you how phase coherence decays with layer phase shifts. It is the kernel that converts local disorder into the full intensity giving both Bragg and diffuse components.

Define

$$g_j \equiv 1 - \tau_j + \tau_j e^{-i\phi}, \qquad S_\ell \equiv \prod_{j=0}^{\ell} g_j. \tag{S38}$$

We wish to compute

$$\langle S_\ell S_{\ell'}^* \rangle = \langle \prod_{j=0}^{\ell} g_j \prod_{j'=0}^{\ell'} g_{j'}^* \rangle. \tag{S39}$$

Since the $g_j$ are i.i.d., expectations of products factor, and $g_j g_j^* = 1$ identically. We split into two cases:

**Case A: $\ell \geq \ell'$.**

$$S_\ell S_{\ell'}^* \;=\; \underbrace{\Big(\prod_{j=0}^{\ell'} g_j\Big)\Big(\prod_{j'=0}^{\ell'} g_{j'}^*\Big)}_{=\,1\ \text{(common layers)}} \underbrace{\Big(\prod_{j=\ell'+1}^{\ell} g_j\Big)}_{\text{extra layers}} \tag{S40}$$

- For $0 \leq j \leq \ell'$, $g_j g_j^* = 1$, so the first two products give 1. - For $j = \ell'+1, \ldots, \ell$, independence implies

$$\Big\langle \prod_{j=\ell'+1}^{\ell} g_j \Big\rangle = \prod_{j=\ell'+1}^{\ell} \langle g_j \rangle = f^{\,\ell-\ell'}. \tag{S41}$$

Hence

$$\langle S_\ell S_{\ell'}^* \rangle = f^{\,\ell-\ell'}, \qquad (\ell \geq \ell'). \tag{S42}$$

**Case B: $\ell < \ell'$.** By an identical argument, swapping roles,

$$\langle S_\ell S_{\ell'}^* \rangle = \big(f^*\big)^{\ell'-\ell}, \qquad (\ell < \ell').$$

**The Solutions are then**

$$\boxed{\langle S_\ell S_{\ell'}^* \rangle = \begin{cases} f^{\,\ell-\ell'}, & \ell \geq \ell', \\ (f^*)^{\ell'-\ell}, & \ell < \ell'. \end{cases}} \tag{S43}$$

### S3.7 Single-sum form of the double series

With $\ell'' = \ell - \ell'$, the double sum becomes

$$\sum_{\ell=0}^{N_z-1} \sum_{\ell'=0}^{N_z-1} f^{|\ell-\ell'|} e^{i(\phi-\psi)(\ell-\ell')} = N_z \sum_{\ell''=-N_z+1}^{N_z-1} f^{|\ell''|} e^{i(\phi-\psi)\ell''}. \tag{S44}$$

Splitting out $\ell'' = 0$ and the $\pm\ell''$ pairs yields

$$N_z \left[ 1 + \sum_{\ell''=1}^{N_z-1} f^{\ell''} \left( e^{i(\phi-\psi)\ell''} + e^{-i(\phi-\psi)\ell''} \right) \right],$$

with

$$\sum_{\ell''=1}^{N_z-1} f^{\ell''} e^{\pm i(\phi-\psi)\ell''} = \frac{f\, e^{\pm i(\phi-\psi)} \left[ 1 - f^{N_z-1} e^{\pm i(\phi-\psi)(N_z-1)} \right]}{1 - f\, e^{\pm i(\phi-\psi)}}.$$

### S3.8 Infinite-crystal limit $(0 \leq f < 1)$

Letting $N_z \to \infty$ gives

$$\sum_{\ell,\ell'} f^{|\ell-\ell'|} e^{i(\phi-\psi)(\ell-\ell')} = N_z \, \frac{1 - f^2}{1 + f^2 - 2f\cos(\phi - \psi)}. \tag{S45}$$

### S3.9 Scattering cross-section

Substituting (S45),

$$\boxed{\frac{d\sigma}{d\Omega} = |F_m|^2 \, \frac{(2\pi)^2}{A_c} \, \delta(\mathbf{Q} - \mathbf{G}_p) \, N_p N_z \, \frac{1 - f^2}{1 + f^2 - 2f\cos(\phi - \psi)}}. \tag{S46}$$

This is identical to the Hendrix–Teller formula upon identifying $\phi^{\mathrm{HT}} = \phi - \psi$ and noting that their parameter $\cos(\phi^1 - \phi^2)$ equals $f$.